

Table 1: Dataset Configurations and Scaling

Dataset & Task	Raw Size	Cleaned / Sampled Size	Train Size	Test Size	Unique Classes / Targets
AFLOW (Space Group)	2,981,873	299,984 (filtered)	239,987	59,997	169 Space Groups
AFLOW (FE & Stability)	2,981,873	109,093 (sampled)	-	-	Formation Energy, Hull Distance, Stability
MP (Space Group)	-	73,704	58,963	14,741	99 Space Groups
MP (Volume per Atom)	-	174,590	-	17,459	Volume per Atom

Table 2: Continuous Property Regression Results (Deep Ensembles)

Metric	Materials Project (MP)	AFLOW
Formation Energy MAE	0.09749 eV/atom	0.47010 eV/atom
Formation Energy STD	0.19880 eV/atom	0.71571 eV/atom
Hull Distance MAE	0.07490 eV/atom	29.57845 eV/atom
Hull Distance STD	0.19987 eV/atom	47.00839 eV/atom
Volume per Atom MAE	0.677 Å ³ /atom	N/A

- **Materials Project (MP):** The **CompositionMTL** model trained on MP data was wrapped in a large loop of **N_ENSEMBLE = 10** independent models. Because the MP dataset size was highly curated, training 10 separate models to establish the standard deviation (uncertainty) bounds was computationally feasible and provided highly robust confidence intervals.
- **AFLOW Dataset:** For the AFLOW Formation Energy and Stability tasks, the ensemble was constrained to **N_ENSEMBLE = 3**. Even after sampling, the AFLOW data processing was computationally heavier, so the ensemble was reduced to 3 independent models with different initialization seeds. Despite the smaller ensemble size, it still effectively produced the $\mu \pm \sigma$ predictions needed for uncertainty-error correlation.

Table 3: Binary Classification Results (Thermodynamic Stability)

Metric	Materials Project (MP)	AFLOW
Accuracy	0.89816	0.89139
F1-Score	0.93057	0.79818
ROC-AUC	0.95567	0.94884

Table 4: Multi-Class Classification Results (Space Group & Crystal System)

Model & Dataset	Top-1 Accuracy	Top-3 Accuracy	Top-5 Accuracy
Random Forest Ensemble (AFLOW)	0.6823	0.9650	0.9751
Random Forest Ensemble (MP - Raw)	0.6549	0.8354	0.8811
Random Forest Ensemble (MP - Re-ranked)	0.7721	0.8940	0.9189
Crystal System Secondary Model (MP)	0.7437	<i>N/A</i>	<i>N/A</i>

Table 5: Probabilistic Uncertainty and Risk-Controlled Screening

Metric / Threshold	Materials Project (MP)	AFLOW
Uncertainty-Error Correlation (Formation Energy)	0.63957	0.53091
Uncertainty-Error Correlation (Volume per Atom)	0.680	N/A
Coverage within $\mu \pm \sigma$ (Volume per Atom)	0.624	N/A
Risk-Controlled MAE at 10% Coverage (Formation Energy)	0.04456 eV/atom	0.0725 eV/atom
Risk-Controlled MAE at 20% Coverage (Formation Energy)	0.04836 eV/atom	0.0873 eV/atom
Risk-Controlled MAE at 30% Coverage (Formation Energy)	0.05131 eV/atom	0.1076 eV/atom
Risk-Controlled MAE at 40% Coverage (Formation Energy)	0.05426 eV/atom	0.1296 eV/atom
Risk-Controlled MAE at 50% Coverage (Formation Energy)	0.05712 eV/atom	0.1605 eV/atom

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